

Introduction Update Review

Source: Update.md / Section: June 21st self-review - Introduction

2. June 21st self-review

Introduction

- 1.2と1.4は内容が近いので、現在の1.3→1.2と1.4を合体したSubsectionとし、合体したセクション内での重複は削除。まずは重複しているものをリストアップ。
- 1.1は一般論としての3Layerの話が長い。現在の説明文はコメントアウトして、圧縮した文章を作成。
- Depite growing recognition- の文章はOCCの一般論の近くに置いて、前後の文章の流れをよくしたい。
- (e.g., through technologies such as-)の文章は長いので簡潔に。
- Dynamic-occupancy-based controlのsubsectionの中のHistorically, tracking personal-level の文章も以前にあったはず。重複はなくす。前の文章の箇所に動かし統合する。
- 同じsubsectionで、This section focuses onは長いレビュー論文のような書き方。簡潔に前後の文章を繋げて導入する。
- その他、Repetitiveな内容はやめて簡潔かつこの論文での目的説明の流れをスムーズに繋がるように、それぞれの文章や段落間での流れが自然になるように。
- 基本的にはOK。Refは論文の足元を固めるものとして残しておきたいので、Refが入っている文章からはその部分の内容を残す。

追加コメント反映後の前提

- ch2-Review.tex は現在使っていないため、重複確認の比較対象から外す。Introduction 内部での重複だけを整理対象にする。
- 引用付きの文は、単に削除せず、引用が支えている主張を圧縮して残す。特に ABW/dynamic occupancy、real-time sensing、multi-occupant OCC、Jung/Ono/Lei への接続は Introduction の根拠として残す。
- ただし、同じ引用・同じ主張が Introduction 内で複数回出る場合は、最も流れが自然な subsection に集約する。

確認結果: Introduction の重複箇所

A. Dynamic occupancy の説明が 1.2 と 1.4 で重複

重複している内容:

- ch1-Intro.tex 45--51: ABW/hybrid work により dynamic occupancy が発生し、人数と occupant composition が変わるという説明。
- ch1-Intro.tex 67--70: flexible work arrangements により occupancy patterns が highly dynamic になり、fixed/static setpoint が合わないという説明。

修正方針:

- 1.2 側に ABW/hybrid work と dynamic occupancy の説明を集約する。
- 1.4 冒頭の Modern flexible work arrangements... から This mismatch... までは削除またはコメントアウトする。
- 1.4 は dynamic occupancy の再説明ではなく、shared-space GCM / control-resolution 問題に直接入る。
- Occupancy fluctuation の根拠として入っている引用は残す。

元の文章:

\subsection{Thermal comfort consideration with rising trend of Activity-based working}

The rise of remote work and \ac{abw} following the COVID-19 pandemic has further complicated thermal comfort control \cite{marzban_review_2023}.

Dynamic working environments are characterized by dynamic occupancy patterns. Individuals move between different workspaces throughout the day, leading to significant fluctuations in occupancy density and distribution within a single zone\cite{motuziene_office_2022,pan_environmental_2025,huang_space-matching_2025,chang_statistical_2013,levene_problem_2020}.

In such spaces, dynamic occupancy should be understood not only as a change in the number of occupants, but also as a change in who is present.

Even when the occupancy count is similar, the thermal preferences represented by the present group may differ as occupants arrive, leave, or move between rooms.

When considering such dynamic occupancy, it becomes extremely challenging to maintain consistent thermal conditions that satisfy all occupants using traditional, static control methods.

A static approach, optimized for a specific occupancy scenario, will inevitably lead to discomfort for some individuals when the actual occupancy deviates from that scenario.

This can result in reduced occupant satisfaction and potentially increased energy consumption as occupants attempt to individually adjust their local environment.

\subsection{Complex occupancy in multi-shared spaces}

Modern flexible work arrangements have led to highly dynamic and variable occupancy patterns in office environments.

Conventional \ac{hvac} systems, which often rely on fixed, static temperature setpoints (e.g., based on the \ac{pmv} model), frequently fail to respond effectively to these fluctuations.

This mismatch can result in both occupant discomfort and unnecessary energy consumption\cite{ono_effects_2022}.

For actual building operation scenarios, both occupancy resolution and comfort model resolution must be considered.

置換後の 1.2 冒頭候補:

\subsection{Dynamic occupancy in activity-based working environments}

The rise of remote and hybrid work has increased the use of \ac{abw} and other flexible office arrangements \cite{marzban_review_2023}.

In such workplaces, occupants move between different spaces throughout the day, causing fluctuations in occupancy density and distribution within a zone \cite{motuziene_office_2022,pan_environmental_2025,huang_space-matching_2025,chang_statistical_2013,levene_problem_2020}.

Dynamic occupancy should therefore be understood not only as a change in the number of people in a room, but also as a change in who is present.

Even when the occupancy count is similar, the thermal preferences represented by the present group may differ as occupants arrive, leave, or move between rooms.

This makes static temperature control less reliable, because a condition optimized for one attendance pattern may not represent another.

->Occupancy fluctuationのRefについても残すこと。

B. Real-time occupancy tracking の背景が 1.3 と 1.5 で重複

重複している内容:

- ch1-Intro.tex 58--64: IoT/sensor technologies により real-time occupancy data が利用可能になったという説明。
- ch1-Intro.tex 111--114: 以前は personal-level occupancy tracking が難しかったが、IoT により可能になり、real-time occupancy の価値を評価する必要があるという説明。

修正方針:

- 1.3 を 1.2 に統合し、real-time sensing の背景は 1 段落に圧縮する。ただし、引用が支える「tracking 技術が使えるようになった」という根拠は残す。
- 1.5 の Historically, tracking personal-level occupancy... は削除し、1.2 統合段落へ移す。
- \autoref{table:OccuTrack} は、Appendix table を有効化するなら Introduction に残してよい。表を使わない場合のみ、文献引用付きの一般文に置き換える。
- Occupancy prediction については今回の主題から外れるため、本文ではコメントアウト候補とする。

元の文章:

\subsection{Treating dynamic occupancy information}

To incorporate occupancy information in \ac{occ} frameworks, two main approaches can be identified: prediction-based methods and real-time sensing.

Many studies have attempted to predict occupancy patterns using historical data and machine learning techniques.

However, as noted in prior research \cite{goyal_occupancy-based_2013}, such prediction methods often

entail high computational costs, may be more expensive than real-time sensing, and may not offer sufficient accuracy or reliability.

On the other hand, recent advances in IoT and sensor technologies have made it increasingly feasible to obtain real-time occupancy data with relatively low cost and high granularity (e.g., through technologies such as passive infrared sensors, CO₂ sensors, and Wi-Fi-based tracking systems) \cite{soleimanijavid_challenges_2024, zafari_survey_2019, brambilla_potential_2021}.

Real-time sensing may enable adaptive control strategies that respond immediately to the presence and number of occupants, which is particularly valuable in shared and flexible workspaces. The availability of real-time data fundamentally shifts the design paradigm of occupant-centric systems from predictive to responsive, thereby enhancing both energy efficiency and thermal comfort performance.

Especially after the COVID-19 and raise of hybrid working, the need of real-time tracking of employee became highly focused.

\autoref{table:OccuTrack} in appendix exhibits examples of Japanese office projects that implement real-time tracking using IoT data.

As such examples show, recent technology enables office management dealing with detailed individual tracking data.

Historically, tracking personal--level occupancy was difficult in typical \ac{bms}; this has changed with IoT systems enabling occupancy/location tracking \cite{soleimanijavid_challenges_2024}.

Thus, \autoref{table:OccuIntegration} underlines the potential of real--time occupancy for \ac{occ} frameworks as an underexplored domain.

Although real-time occupancy data are increasingly available, their practical value depends on whether the additional occupant-composition information produces enough comfort benefit to justify the sensing, integration, and control burden.

Therefore, this study evaluates both the comfort improvement and the operational implications of using higher-resolution \ac{p-gcm}.

置換候補:

Recent IoT and building-management technologies have made room- or occupant-level occupancy data more accessible through sources such as PIR, CO₂, and Wi-Fi-based sensing \cite{soleimanijavid_challenges_2024, zafari_survey_2019, brambilla_potential_2021}.

\autoref{table:OccuTrack} provides examples of office projects that use IoT data for real-time occupancy tracking.

These data can support responsive \ac{occ}, but their practical value depends on whether the additional occupant-composition information produces enough comfort benefit to justify the sensing, integration, privacy, and control burden.

->原則OK。さらに、今回はOccupancy predictionについて触れる必要はないため、該当箇所はコメントアウト。

C. Shared-space OCC / GCM 問題が 1.1, 1.4, 1.5, 1.6 に分散

重複している内容:

- ch1-Intro.tex 37--39: shared office では isolated PCM だけではなく room-level GCM が必要。
- ch1-Intro.tex 72--80: shared spaces では multiple occupants の conflicting preferences と comfort/control resolution の問題がある。
- ch1-Intro.tex 83--89, 105--109: static occupant configuration を仮定する既存研究が多く、Lei et al. は例外だが computationally demanding。
- ch1-Intro.tex 117--124: focus of research で同じ research gap を再度説明。

修正方針:

- 1.1 は OCC の一般定義と「shared-space では GCM が必要」までに圧縮する。
- 1.4 と 1.5 を統合して Shared-space GCM resolution under dynamic occupancy とする。
- 1.6 は research question と contribution だけにし、gap の再説明は避ける。
- ch2-Review.tex は使っていないため、table:OccuIntegration を Introduction から削除する必要はない。ただし、Introduction の流れを重くする場合は、table 本体ではなく引用付きの短い文章として残す。
- table:OccuIntegration は Introduction の根拠として残す。
- ただし、table の説明は短くし、既存研究の static/dynamic adaptation の比較を支える位置づけにする。

元の文章:

In shared office spaces, however, \ac{occ} cannot rely only on isolated \ac{pcm}, because a single room-level \ac{hvac} setpoint must represent multiple occupants at the same time. This makes the construction of a \ac{gcm}, which aggregates individual comfort information into a room-level representation, a central design issue for shared-space \ac{occ}. The key question is not only how accurately individual \ac{pcm} predict comfort, but also which occupant set should be represented when the room-level control decision is made.

Effective \ac{occ} systems require determining proper control resolution and comfort model resolution depending on the room/building usage. Existing \ac{occ} research has largely focused on developing and validating \ac{pcm} and control algorithms in single-occupancy settings \cite{park_critical_2019, jung_human---loop_2019}. While these studies demonstrate the potential of \ac{occ}, few have tried to address the complexities of shared spaces. Shared workspaces present a significant challenge for \ac{occ}, as multiple occupants with potentially conflicting thermal preferences must be accommodated simultaneously \cite{xie_review_2020}. Simply averaging preferences or applying a single \ac{pcm} is unlikely to satisfy everyone, and may even exacerbate discomfort for some individuals. \mycite{jung_energy_2020} investigated how the number of occupants in a zone affects \ac{hvac} system performance, considering both comfort and energy consumption. They found that both performances could decrease as the number of occupants increases, and results could differ depending on the methods used to integrate individual comfort models \cite{jung_energy_2020}. \mycite{ono_effects_2022} investigated how the resolution of thermal comfort models and \ac{hvac} control settings should match. They found that mismatches between comfort model resolution and control resolution, especially when using detailed comfort models with simpler \ac{hvac} systems, may reduce effectiveness and decrease occupant satisfaction \cite{ono_effects_2022}.

As noted in reviews \cite{xie_review_2020}, integrating diverse preferences across multiple occupants remains underexplored. This section focuses on how comfort models are incorporated into control. In some studies on the effectiveness of \ac{occ}, such as \cite{jung_energy_2020}, occupancy is often assumed to be static. However, in today's dynamic working environments--where hybrid working, combining remote and office-based work, has become commonplace\cite{redmond_rise_nodate} --this static assumption may underestimate the impact of occupancy characteristics. \autoref{table:OccuIntegration} compares field studies that integrate multiple \ac{pcm} within the same control zone. The "Occupancy Adaptation" column indicates each framework's ability to handle different occupancy patterns; "dynamic" adjusts to real--time combinations, while "static" assumes fixed configurations. Only \cite{lei_practical_2022} considers transitions among combinations by pre--training multiple RL models for frequent patterns--data--and compute--intensive.

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\begin{table}[pos=!htbp]
  \centering
  \input{pic/ch2-Review/ModelIntegration.tex}
  \caption{Field OCC studies integrating multiple personal comfort models}
  \label{table:OccuIntegration}
\end{table}
```

While these studies have investigated multi-occupancy scenarios, a significant limitation is that most assume a static occupant attendance configuration, meaning they do not account for occupants arriving and leaving dynamically \cite{lee_implementation_2019, li_indoor_2019, li_personalized_2017, li_indoor_2023, aguiler_a_thermal_2019, li_experimental_2021}.

A notable exception is the work by Lei et al.~(2022), which addressed dynamic occupancy in a shared-lab office \cite{lei_practical_2022}. They employed multiple reinforcement learning models tailored to the most common combinations of occupants. However, this approach is computationally demanding as it requires maintaining and training numerous models.

統合後の subsection 候補:

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\subsection{Shared-space GCM resolution under dynamic occupancy}
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In shared offices, a single room-level \ac{hvac} setpoint must represent multiple occupants with potentially different thermal preferences. Existing \ac{occ} research has largely focused on developing \ac{pcm} and control algorithms in single-occupancy settings, while shared spaces remain less explored \cite{park_critical_2019, jung_human---loop_2019, xie_review_2020}. \mycite{jung_energy_2020} showed that occupant number can affect both comfort and energy performance, depending on how individual comfort models are integrated. \mycite{ono_effects_2022} further showed that mismatches between comfort-model resolution and control resolution may reduce control effectiveness. These studies indicate that shared-space \ac{occ} requires attention to both comfort aggregation and room-level control resolution. \autoref{table:OccuIntegration} compares field studies that integrate multiple \ac{pcm} within the same control zone. Most existing multi-occupant studies assume static occupant attendance configurations and do not account for occupants arriving and leaving dynamically \cite{lee_implementation_2019, li_indoor_2019}

,li_personalized_2017,li_indoor_2023,aguilera_thermal_2019,li_experimental_2021}.
Lei et al.~\cite{lei_practical_2022} addressed dynamic occupancy using multiple reinforcement-learning models for frequent occupant combinations, but such an approach can be data- and compute-intensive. Therefore, a simpler practical question remains: when is static, daily, or real-time aggregation of individual comfort information sufficient for room-level control?

->方向性は良いが、Table:OccuIntegrationへの引用は残しておく。reviewのtexのみ不採用としたが、引用Tableのtexはこのまま採用。

D. This section focuses on... は削除

対象:

As noted in reviews \cite{xie_review_2020}, integrating diverse preferences across multiple occupants remains underexplored.
This section focuses on how comfort models are incorporated into control.

置換候補:

For room-level \ac{occ}, the unresolved issue is how individual comfort information should be aggregated when the set of occupants changes over time.

->OK

Introduction の推奨構成

現在の subsection を以下に再構成する。

1. Occupant-centric control and shared-space comfort aggregation
 - OCC の定義を1段落に圧縮。
 - 3-layer 説明はコメントアウトし、sensing/learning/control は1文だけにする。
 - Despite growing recognition... はこの subsection の最後に置く。
2. Dynamic occupancy in activity-based working environments
 - 現在の 1.2 と 1.3 を統合。
 - ABW/hybrid work、occupant composition、real-time sensing、practical burden をここで説明。
3. Shared-space GCM resolution under dynamic occupancy
 - 現在の 1.4 と 1.5 を統合。
 - Jung, Ono, Lei への接続をここに置く。
 - \autoref{table:OccuIntegration} は、Introduction の根拠として必要なら残す。本文が重くなる場合は table を出さず、Jung/Ono/Lai を引用した短い文章に圧縮する。
4. Focus of the research
 - 研究目的、SGCM/DGCM/RT-GCM、評価指標だけに絞る。

1.1 圧縮案

現在の 3-layer 説明 (ch1-Intro.tex 23--36) は長いためコメントアウトし、以下に置換する。

元の文章:

In recent years, \ac{occ} has emerged as a promising alternative \ac{hvac} strategy that decides control actions by considering individual thermal preferences \cite{soleimanijavid_challenges_2024}. In definition, \ac{occ} leverages sensor data and \ac{pcm} to adjust \ac{hvac} operation, potentially improving both thermal comfort and energy efficiency as examined by existing studies \cite{xie_review_2020}. In this context, IEA EBC Annex 79 defines \ac{occ} as an "occupant-in-the-loop" approach that seeks to provide optimal and personalized conditions rather than imposing fixed settings\cite{wagner_international_2024}.

As some existing research has exhibited, the general \ac{occ} framework is composed of a combination of

three components, namely "Sensing," "Learning," and "Control" layers \cite{soleimanijavid_challenges_2024,xie_review_2020}.

In the sensing layer, information is gathered from various sources, such as environmental data, occupants' comfort sensation, or occupants' attendance.

In the learning layer, prediction models on environmental preference or occupancy are developed.

In the control layer, considering these occupancy and preference information, control actions are decided based on tailored algorithms.

Among these components, the thermal preference model and its integration into the control layer have a significant impact on occupant comfort, hence they should be carefully considered as a framework.

In recent years, \ac{occ} has emerged as a promising \ac{hvac} strategy that uses sensing, learning, and control functions to reflect occupant needs in building operation \cite{soleimanijavid_challenges_2024,xie_review_2020}.

IEA EBC Annex 79 defines \ac{occ} as an ``occupant-in-the-loop'' approach that seeks to provide optimal and personalized conditions rather than imposing fixed settings \cite{wagner_international_2024}.

In shared office spaces, however, \ac{occ} cannot rely only on isolated \ac{pcm}, because a single room-level \ac{hvac} setpoint must represent multiple occupants at the same time.

This makes \ac{gcm} construction, and especially the choice of which occupants are represented, a central design issue for shared-space \ac{occ}.

Despite growing recognition in the industry and active framework development in simulations, real-world \ac{occ} applications remain limited \cite{soleimanijavid_challenges_2024}.

Focus of research の圧縮案

現在の ch1-Intro.tex 117--124 は gap の説明を繰り返しているため、以下のように目的へ直接接続する。

元の文章:

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\subsection{Focus of the research}
As highlighted in the background, the practical application of \ac{occ} in shared office spaces is still limited by the common assumption of static occupancy.
However, occupant composition in a room can change across longer-term, daily, and real-time timescales. This raises a practical question: at what occupancy conditions does higher-resolution aggregation of individual comfort information become worthwhile for room-level control?
This study asks when higher-resolution aggregation of individual comfort information improves predicted comfort in shared office spaces, and what operational burden is introduced by real-time occupant-composition updates.

To address this question, this study focuses on three group comfort models that differ in the timescale at which individual \ac{pcm} are aggregated: a static model based on regular room members, a daily model based on a day's attendees, and a real-time model based on the occupants currently present.
Building on prior work that investigated the impact of group size and control granularity \cite{ono_effects_2022,jung_energy_2020}, this study introduces temporal occupancy variability as a key dimension of analysis.
Using field data from real office spaces, the study investigates how the relative benefit of these models changes with occupancy conditions such as group size, occupancy variability, and user turnover.

This study evaluates when higher-resolution aggregation of individual comfort information improves predicted comfort in shared offices, and what operational burden is introduced by real-time occupant-composition updates.
Three \ac{gcm} are compared: a \ac{sgcm} based on regular room members, a \ac{dgcm} based on daily attendees, and a \ac{rtgcm} based on occupants present at each control decision window.
Building on prior work on group size and control granularity \cite{ono_effects_2022,jung_energy_2020}, the analysis relates model performance to subgroup size, mean occupancy, occupancy variability, and \ac{utr}.
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既存の Specifically, the analysis compares... はそのまま残してよい。

その他の Repetitive / 整合性問題

1. table:OccuIntegration が Introduction と Review で重複

- ch1-Intro.tex 91--95 と ch2-Review.tex 136--140 で同じ table と label を使用している。
- ただし ch2-Review.tex は現在使っていないため、これは現時点ではコンパイル上の優先問題ではない。
- Introduction の論旨を支える根拠として table が有効なら、Introduction 側に残してよい。
- 残す場合は、table の前後説明を短くし、Occupancy Adaptation column の細かい説明は削る。
- table を出さない場合でも、Jung/Ono/Lai と static/dynamic adaptation の論点は引用付き文章として残す。

2. fig:Upd-Field label が複数箇所で重複

- ch2-Review.tex 143--147: dynamic occupancy schematic。
- ch2-Review.tex 173--177: model update figure。
- ch1-Intro.tex 98--102 はコメントアウトされているが、復活させるとさらに重複する。

修正案:

- ch2-Review.tex は使っていないため、現時点では Introduction 内で復活させる図だけ注意する。
- Introduction のコメントアウト図を復活させる場合は fig:dynamic_occupancy_schematic_intro などに変更する。
- 図を使わないならコメントアウトブロックごと削除してよい。

3. Introduction 内の dynamic-occupancy review がやや重い

- ch2-Review.tex は前提にしないため、Introduction 内に必要な引用と根拠は残す。
- ただし、table description や The "Occupancy Adaptation" column... のような表の読み方説明は Introduction では長い。
- Introduction は「既存研究は static multi-occupant assumption が多い」「Jung/Ono は group size や control/comfort resolution の根拠になる」「Lei et al. は動的 occupancy を扱う例外だが重い」「本研究は simple GCM resolution comparison に絞る」の 4 点を残す。